Solution to Exercise 03 — Analysis on Manifolds, 80416

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Question 1

For every domain $\Omega \subseteq \mathbb{R}^n$ and vector field $F : \Omega \to \mathbb{R}^n$ we will find an explicit function $u : \Omega \to \mathbb{R}$ which is a potential for F.

Subquestion 1

$\Omega = \mathbb{R}^3,\ F(x, y, z) = (y + z, x + z, x + y)$.

Solution: We want u such that

$$\nabla u = F = (y + z, x + z, x + y)$$

That is

$$\frac{\partial}{\partial x} u = y + z,\ \frac{\partial}{\partial y} u = x + z,\ \frac{\partial}{\partial z} u = x + y$$

We start by integrate with respect to x (so y, z are constants)

$$u(x, y, z) = \int (y + z)\,\mathrm{d}x = x(y + z) + C(y, z) = xy + xz + C(y, z)$$

Now we differentiate with respect to y

$$\frac{\partial}{\partial y} u = x + \frac{\partial}{\partial y} C(y, z)$$

We need

$$x + \frac{\partial}{\partial y} C(y, z) = x + z$$

Hence

$$C(y, z) = z$$

We integrate with respect to y and get that $C(y, z) = yz + D(z)$ and we have $u(x, y, z) = xy + xz + yz + D(z)$.
We differentiate with respect to z

$$\frac{\partial}{\partial z} u = x + y + \frac{\partial}{\partial z} D(z)$$

We need

$$x + y + \frac{\partial}{\partial z} D(z) = x + y$$

Hence $\frac{\partial}{\partial z} D(z) = 0$ which means that $D(z)$ is a constant c and the wanted u is given by

$$u(x, y, z) = xy + xz + yz + c$$

Sanity check:

$$\nabla u = (y + z, x + z, x + y) = F$$

$\square$

Subquestion 2

$\Omega = \mathbb{R}^2 \setminus \{0\},\ F(x,y) = \left(\frac{x}{x^2+y^2}, \frac{y}{x^2+y^2}\right)$.

Solution: We want u such that

$$\nabla u = F = \left(\frac{x}{x^2+y^2}, \frac{y}{x^2+y^2}\right)$$

That is

$$\frac{\partial}{\partial x} u = \frac{x}{x^2+y^2},\ \frac{\partial}{\partial y} u = \frac{y}{x^2+y^2},$$

Let We start by integrate with respect to x

$$u(x,y) = \int \frac{x}{x^2+y^2}\,\mathrm{d}x \underset{\substack{u=x^2+y^2 \\ \mathrm{d}u=2x\,\mathrm{d}x}}{=} \int \frac{1}{2u}\,\mathrm{d}u = \frac{1}{2}\ln(|u|) = \frac{\ln(x^2+y^2)}{2} + C(y)$$

Now we differentiate with respect to y

$$\frac{\partial}{\partial y} u = \frac{y}{x^2+y^2} + \frac{\partial}{\partial y} C(y)$$

We need

$$\frac{\partial}{\partial y} u = \frac{y}{x^2+y^2}$$

Hence $\frac{\partial}{\partial y} C(y) = 0$ which means that $C(y) = c$ a constant hence the wanted u is given by

$$u(x,y) = \frac{\ln(x^2+y^2)}{2} + c$$

Sanity check:

$$\nabla u = \left(\frac{x}{x^2+y^2}, \frac{y}{x^2+y^2}\right)$$

$\square$

Question 2

Denote $\Omega = \mathbb{R}^3 \setminus \{(0,0,z) \mid z \in \mathbb{R}\}$. Let $F : \Omega \to \mathbb{R}^3$ be a vector field defined by $F(x,y,z) = \left(\frac{x}{x^2+y^2}, \frac{y}{x^2+y^2}, z\right)$.

Subquestion 1

We will show that F is conservative.

Proof: First we notice that $\Omega = \mathbb{R}^3 \setminus \{(0,0,z) \mid z \in \mathbb{R}\}$ is not simply connected so the best way to show that F is conservative is by using a theorem from the leacture: F is conservative if and only if $F = \nabla u$ for some $u : \Omega \to \mathbb{R}$ so we will construct a proper u, that is

$$\frac{\partial}{\partial x}u = \frac{x}{x^2+y^2}, \ \frac{\partial}{\partial y}u = \frac{y}{x^2+y^2}, \ \frac{\partial}{\partial z}y = z$$

We start by integrate with respect to x

$$u(x,y,z) = \int \left(\frac{x}{x^2+y^2}\right) \mathrm{d}x = \frac{\ln(x^2+y^2)}{2} + C(y,z)$$

We differentiate with respect to y

$$\frac{\partial}{\partial y}u = \frac{y}{x^2+y^2} + \frac{\partial}{\partial y}C(y,z)$$

We need

$$\frac{\partial}{\partial y}u = \frac{y}{x^2+y^2}$$

Hence

$$\frac{\partial}{\partial y}C(y,z) = 0 \Longrightarrow C(y,z) = C(z)$$

So far we have

$$u(x,y,z) = \frac{\ln(x^2+y^2)}{2} + C(z)$$

Since

$$\frac{\partial}{\partial z}u = z = \frac{\partial}{\partial z}C(z)$$

We integrate with respect to z and get that

$$C(z) = \frac{z^2}{2} + c$$

We attain that u is given by

$$u(x,y,z) = \frac{\ln(x^2+y^2)}{2} + \frac{z^2}{2} + c$$

hence $F = \nabla u$ and F is conservative. $\square$

Subquestion 2

We will calculate $\int_\gamma \vec{F}\,\mathrm{d}\vec{\ell}$ where $\gamma(t) = \big(\cos(t), \sin(t), t^2\big)$ and $t \in [0, 2\pi]$.

Solution:

Reminder: Let $\gamma : [a, b] \to \mathbb{R}$ and $\vec{X}$ a vector field then

$$\int_\gamma \vec{X}\,\mathrm{d}\vec{\ell} = \int_a^b \vec{X}(\gamma(t)) \cdot \dot{\gamma}(t)\,\mathrm{d}t$$

We compute each needed part:

$$\dot{\gamma}(t) = (-\sin(t), \cos(t), 2t)$$

$$F(\gamma(t)) = \left(\frac{\cos(t)}{\cos^2(t) + \sin^2(t)}, \frac{\sin(t)}{\cos^2(t) + \sin^2(t)}, t^2\right) \underset{\cos^2(t)+\sin^2(t)=1}{=} \big(\cos(t), \sin(t), t^2\big)$$

$$F(\gamma(t)) \cdot \dot{\gamma}(t) = \big(\cos(t), \sin(t), t^2\big) \cdot (-\sin(t), \cos(t), 2t) = -\sin(t)\cos(t) + \sin(t)\cos(t) + 2t^3 = 2t^3$$

Finally

$$\int_\gamma \vec{F}\,\mathrm{d}\vec{\ell} = \int_0^{2\pi} 2t^3\,\mathrm{d}t = \left[\frac{t^4}{2}\right]_{t=0}^{t=2\pi} \frac{(2\pi)^4}{2} = 8\pi^4$$

$\square$

Question 3

Let $U \subseteq \mathbb{R}^n$ be an open set and let $\gamma_0, \gamma_1 : [a,b] \to U$ be two curves such that $\gamma_0(a) = \gamma_1(a),\ \gamma_0(b) = \gamma_1(b)$.
We say that γ_0, γ_1 are homotopic in U if there exists a smooth map $H : [a,b] \times [0,1] \to U$ such that
1. $\forall t \in [a,b],\ H(t,0) = \gamma_0(t),\ H(t,1) = \gamma_1(t)$
2. $\forall s \in [0,1],\ H(a,s) = \gamma_1(a),\ H(b,s) = \gamma_1(b)$

Let Ω be a domain and let $F : \Omega \to \mathbb{R}^n$ be a locally conservative vector field.

Subquestion 1

Let $\gamma_0, \gamma_1 : [a,b] \to \Omega$ such that $\gamma_0(a) = \gamma_1(a),\ \gamma_0(b) = \gamma_1(b)$ and we assume that $\gamma_0,\ \gamma_1$ are homotopic. We will prove that

$$\int_{\gamma_0} \vec{F}\,\mathrm{d}\vec{\ell} = \int_{\gamma_1} \vec{F}\,\mathrm{d}\vec{\ell}$$

Proof: In other words, homotopic means that one curve can be smoothly deformed into the other curve.
We denote $H : [a,b] \times [0,1] \to U$ be the homotopy map and $\gamma_s(t) = H(t,s)$ so that for every $s \in [0,1]$ the curve $\gamma_s : [a,b] \to U$ is a curve from $\gamma_0(a) = \gamma_1(a)$ to $\gamma_0(b) = \gamma_1(b)$ which means that $H(t,s)$ is smoothly deforms the starting curve γ_0 into the final curve γ_1 as the parameter s moves from 0 to 1.
Let $I(s) = \int_{\gamma_s} \vec{F}\,\mathrm{d}\vec{\ell}$ (where $I(0)$ represents the integral along the γ_0 curves and $I(1)$ the integral along the γ_1 curve) and if we show that $I(s)$ is a constant we are done (since if $I(s)$ is a constant for all values of s then it's value does not change during the deformation meaning that $I(0) = I(1)$ hence $\int_{\gamma_0} \vec{F}\,\mathrm{d}\vec{\ell} = \int_{\gamma_1} \vec{F}\,\mathrm{d}\vec{\ell}$).
We denote $\vec{F} = (F_1, ..., F_n)$ and $H(t,s) = (x_1(t,s), ..., x_n(t,s))$ and we need to compute

$$I(s) = \int_{\gamma_s} \vec{F}\,\mathrm{d}\vec{\ell} = \int_a^b \vec{F}(H(t,s)) \cdot \frac{\partial H}{\partial t}(t,s)\,\mathrm{d}t$$

The integrand in the right hand side is smooth and continuously differentiable with respect to s hence I is differentiable so it will suffice to show that $I'(s) = 0$

$$\begin{aligned}
I'(s) &= \frac{\mathrm{d}}{\mathrm{d}s} \int_a^b \vec{F}(H(t,s)) \cdot \frac{\partial H}{\partial t}(t,s)\,\mathrm{d}t \\
&\underset{(1)}{=} \int_a^b \frac{\mathrm{d}}{\mathrm{d}s} \left(\vec{F}(H(t,s)) \cdot \frac{\partial H}{\partial t}(t,s) \right) \mathrm{d}t \\
&\underset{(2)}{=} \int_a^b \frac{\mathrm{d}}{\mathrm{d}s} \left(\sum_{i=1}^n F_i(x_1(t,s), ..., x_n(t,s)) \frac{\partial x_i}{\partial t}(t,s) \right) \mathrm{d}t \\
&\underset{(3)}{=} \int_a^b \sum_{i=1}^n \sum_{j=1}^n \frac{\partial F_i}{\partial x_j}(x_1(t,s), ..., x_n(t,s)) \frac{\partial x_j}{\partial s}(t,s) \frac{\partial x_i}{\partial t}(t,s) + \sum_{i=1}^n F_i(x_1(t,s), ..., x_n(t,s)) \frac{\partial^2 x_i}{\partial t \partial s}(t,s)\,\mathrm{d}t \\
&\underset{(4)}{=} \int_a^b \sum_{i=1}^n \sum_{j=1}^n \frac{\partial F_j}{\partial x_i}(x_1(t,s), ..., x_n(t,s)) \frac{\partial x_j}{\partial s}(t,s) \frac{\partial x_i}{\partial t}(t,s) + \sum_{i=1}^n F_i(x_1(t,s), ..., x_n(t,s)) \frac{\partial^2 x_i}{\partial t \partial s}(t,s)\,\mathrm{d}t \\
&\underset{(5)}{=} \int_a^b \frac{\mathrm{d}}{\mathrm{d}t} \left(\sum_{j=1}^n F_j(x_1(t,s), ..., x_n(t,s)) \frac{\partial x_j}{\partial s}(t,s) \right) \mathrm{d}t \\
&\underset{(6)}{=} \left[\sum_{j=1}^n F_j(x_1(t,s), ..., x_n(t,s)) \frac{\partial x_j}{\partial s}(t,s) \right]_a^b \\
&\underset{(7)}{=} \sum_{j=1}^n F_j(H(t,s)) \underbrace{\frac{\partial x_j}{\mathrm{d}s}(b,s)}_{=0} - \sum_{j=1}^n F_j(H(t,s)) \underbrace{\frac{\partial x_j}{\mathrm{d}s}(a,s)}_{=0} = 0
\end{aligned}$$

Explanations for each step: (1) is due to the fact that both are smooth in both variables so we can differentiate under the integral sign (change differentiation and integral order), (2) is opening definitions, (3) is the Chain and Leibniz Integral Rule from the recitation, (4) is due to F being a locally conservative vector field, (5) is the Chain Rule again with respect to t since in the previous step the roles of t and s were switched, (6) is the Fundamental Theorem of Calculus and (7) is since the end points does not moves so it's devriative is zero and we can change s as we want but $x_j(a,s) \equiv p$ for p the matching end point which means that this is a constant curve with respect to s. $\square$

Subquestion 2

We will prove that if every two curves $\gamma_0, \gamma_1 : [a,b] \to \Omega$ such that $\gamma_0(a) = \gamma_1(a),\ \gamma_0(b) = \gamma_1(b)$ are homotopic then F is conservative (we call a domain Ω that satisfies this property **simply connected**).

Proof: From the previous part we have

$$\int_{\gamma_0} \vec{F}\,\mathrm{d}\vec{\ell} = \int_{\gamma_1} \vec{F}\,\mathrm{d}\vec{\ell}$$

So we have path independence.
Let $x_0 \in \Omega$ and γ is any curve from x_0 to x and we define

$$u(x) = \int_{\gamma} \vec{F}\,\mathrm{d}\vec{\ell}$$

$u(x)$ is well defined from the previous part and we want to show that $u(x)$ is a potential so we prove $\nabla u = F$.
For the i-th partial derivative we take a small increment h in the i-th direction then where γ_0 is a curve from x_0 to $x + he_i$

$$(\star)\ u(x + he_i) = \int_{\gamma_0} \vec{F}\,\mathrm{d}\vec{\ell}$$

We use the path path independence and we can write

$$u(x + he_i) = u(x) + \int_{\gamma_1} \vec{F}\,\mathrm{d}\vec{\ell}$$

Where γ_1 is the curve from x to $x + he_i$ given by $\gamma_1(t) = x + the_i$ for $t \in [0,1]$ then $\gamma_1'(t) = he_i$ so on the right side we have

$$\int_0^1 F(\gamma_1(t)) \cdot (he_i)\,\mathrm{d}t$$

Combine the above and $(\star)$ becomes

$$\frac{u(x + he_i) - u(x)}{h} = \int_0^1 F(\gamma_1(t)) \cdot (e_i)\,\mathrm{d}t$$

We take limit as $h \to 0$ and we attain that $F(\gamma_1(t)) \to F(x)$ hence (from the integration is over constant)

$$\frac{\partial u}{\partial x_i}(x) = F(x) \cdot e_i = F_i(x)$$

This holds for every component hence $\partial u = F$ and F is conservative. $\square$

Question 4

Let $f, g : [a, b] \to \mathbb{R}$ be C^1 functions such that $f \leq g$. We Denote $A = \{(x, y) \in \mathbb{R}^2 : x \in [a, b], y \in [f(x), g(x)]\}$.
For $h \in \{f, g\}$ we define $\gamma_h : [a, b] \to \mathbb{R}^2$ by $\gamma_h(t) = (t, h(t))$ and for $c \in \{a, b\}$ we define $\delta_c : [f(c), g(c)] \to \mathbb{R}^2$ by $\delta_c(t) = (c, t)$.
Let Ω be a neighborhood of A and let $F = (P, Q) : \Omega \to \mathbb{R}^2$ be a smooth vector field.

Subquestion 1

We will show that

$$\int_A \frac{\partial P}{\partial y}(x, y)\,\mathrm{d}x\,\mathrm{d}y = \int_{\gamma_g} (P, 0)\,\mathrm{d}\ell - \int_{\gamma_f} (P, 0)\,\mathrm{d}\ell$$

Proof: By the definition of A we have x goes from a to b and for each fixed x, y goes from $f(x)$ to $g(x)$ which means we slice vertically so integration over A means $\mathrm{d}y\,\mathrm{d}x$ if we move to iterated integrals.
Hence on the left side we have

$$\int_A \frac{\partial P}{\partial y}(x, y)\,\mathrm{d}x\,\mathrm{d}y = \int_a^b \int_{f(x)}^{g(x)} \frac{\partial P}{\partial y}(x, y)\,\mathrm{d}y\,\mathrm{d}x \underset{(\star)}{=} \int_a^b P(x, g(x)) - P(x, f(x))\,\mathrm{d}x = \int_a^b P(x, g(x))\,\mathrm{d}x - \int_a^b P(x, f(x))\,\mathrm{d}x$$

Where $(\star)$ is since that for a fixed x we treat $P(x, y)$ as a function of y and use the Fundamental Theorem of Calculus in y.
For any curve $\gamma(t) = (x(t), y(t))$ and vector field $F = (F_1, F_2)$ we have from definition

$$\int_\gamma F\,\mathrm{d}\ell = \int_a^b F(x(t), y(t))(x'(t), y'(t))\,\mathrm{d}t \underset{F=(P,0)}{=} \int_a^b P(x(t), y(t))x'(t)\,\mathrm{d}t$$

So on the right side we have

$$\int_{\gamma_g} (P, 0)\,\mathrm{d}\ell - \int_{\gamma_f} (P, 0)\,\mathrm{d}\ell = \int_a^b P(t, g(t)) \cdot 1\,\mathrm{d}t - \int_a^b P(t, f(t)) \cdot 1\,\mathrm{d}t$$

With a change of variable $\mathrm{d}x = \mathrm{d}t$ we got that the left side and right side are equal. $\square$

Subquestion 2

We will show that

$$\int_A \frac{\partial Q}{\partial x}(x, y)\,\mathrm{d}x\,\mathrm{d}y = \int_{\delta_b} F\,\mathrm{d}\ell - \int_{\delta_a} F\,\mathrm{d}\ell + \int_{\gamma_f} (0, Q)\,\mathrm{d}\ell - \int_{\gamma_g} (0, Q)\,\mathrm{d}\ell$$

Proof: From the same reasoing as above about A

$$\int_A \frac{\partial Q}{\partial x}(x, y)\,\mathrm{d}x\,\mathrm{d}y = \int_a^b \int_{f(x)}^{g(x)} \frac{\partial Q}{\partial x}(x, y)\,\mathrm{d}y\,\mathrm{d}x$$

We can't use the Fundamental Theorem of Calculus so instead we will use Leibniz Integration Rule from the recitation (since all the terms are applied) hence

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\int_{f(x)}^{g(x)} Q(x, y)\,\mathrm{d}y\right) = \int_{f(x)}^{g(x)} \frac{\partial Q}{\partial x}(x, y)\,\mathrm{d}y + Q(x, g(x))g'(x) - Q(x, f(x))f'(x)$$

So the inner integral from above becomes

$$\int_{f(x)}^{g(x)} \frac{\partial Q}{\partial x}(x, y)\,\mathrm{d}y = \frac{\mathrm{d}}{\mathrm{d}x}\left(\int_{f(x)}^{g(x)} Q(x, y)\,\mathrm{d}y\right) - Q(x, g(x))g'(x) + Q(x, f(x))f'(x)$$

Hence

$$\int_a^b \int_{f(x)}^{g(x)} \frac{\partial Q}{\partial x}(x,y)\,\mathrm{d}y\,\mathrm{d}x = \int_a^b \frac{\mathrm{d}}{\mathrm{d}x}\left(\int_{f(x)}^{g(x)} Q(x,y)\,\mathrm{d}y\right)\mathrm{d}x - \int_a^b Q(x,g(x))g'(x)\,\mathrm{d}x + \int_a^b Q(x,f(x))f'(x)\,\mathrm{d}x$$

By the Fundamental Theorem of Calculus we have (since $\int_a^b \frac{\mathrm{d}}{\mathrm{d}x}H(x)\,\mathrm{d}x = H(b) - H(a)$ so with $H(x) = \int_{f(x)}^{g(x)} Q(x,y)\,\mathrm{d}y$)

$$\int_a^b \frac{\mathrm{d}}{\mathrm{d}x}\left(\int_{f(x)}^{g(x)} Q(x,y)\,\mathrm{d}y\right)\mathrm{d}x = \int_{f(b)}^{g(b)} Q(b,y)\,\mathrm{d}y - \int_{f(a)}^{g(a)} Q(a,y)\,\mathrm{d}y$$

So far we have

$$\int_a^b \int_{f(x)}^{g(x)} \frac{\partial Q}{\partial x}(x,y)\,\mathrm{d}y\,\mathrm{d}x = \int_{f(b)}^{g(b)} Q(b,y)\,\mathrm{d}y - \int_{f(a)}^{g(a)} Q(a,y)\,\mathrm{d}y - \int_a^b Q(x,g(x))g'(x)\,\mathrm{d}x + \int_a^b Q(x,f(x))f'(x)\,\mathrm{d}x$$

We split each component on the right side and compute

$$\begin{aligned}
\int_{\delta_b} F\,\mathrm{d}\ell &= \int_{f(b)}^{g(b)} (P(b,t),Q(b,t))\cdot(0,1)\,\mathrm{d}t = \int_{f(b)}^{g(b)} Q(b,t)\,\mathrm{d}t \\
\int_{\delta_a} F\,\mathrm{d}\ell &= \int_{f(a)}^{g(a)} (P(a,t),Q(a,t))\cdot(0,1)\,\mathrm{d}t = \int_{f(a)}^{g(a)} Q(a,t)\,\mathrm{d}t \\
\int_{\gamma_g} F\,\mathrm{d}\ell &= \int_a^b (0,Q(t,g(t)))\cdot(1,g'(t))\,\mathrm{d}t = \int_a^b Q(t,g(t))g'(t)\,\mathrm{d}t \\
\int_{\gamma_f} F\,\mathrm{d}\ell &= \int_a^b (0,Q(t,f(t)))\cdot(1,f'(t))\,\mathrm{d}t = \int_a^b Q(t,f(t))f'(t)\,\mathrm{d}t
\end{aligned}$$

Combine them and on the right side we have

$$\int_{f(b)}^{g(b)} Q(b,t)\,\mathrm{d}t - \int_{f(a)}^{g(a)} Q(a,t)\,\mathrm{d}t + \int_a^b Q(t,f(t))f'(t)\,\mathrm{d}t - \int_a^b Q(t,g(t))g'(t)\,\mathrm{d}t$$

With another change of variable $\mathrm{d}t = \mathrm{d}y$ we got that the left side and right side are equal. □

Subquestion 3

We will conclude that

$$\int_A \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\,\mathrm{d}x\,\mathrm{d}y = \int_{\delta_b} F\,\mathrm{d}\ell - \int_{\gamma_g} F\,\mathrm{d}\ell - \int_{\delta_a} F\,\mathrm{d}\ell + \int_{\gamma_f} F\,\mathrm{d}\ell$$

Proof: On the left side with the linearity of the integral we have

$$\begin{aligned}
\int_A \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\,\mathrm{d}x\,\mathrm{d}y &= \int_{\delta_b} F\,\mathrm{d}\ell - \int_{\delta_a} F\,\mathrm{d}\ell + \int_{\gamma_f} (0,Q)\,\mathrm{d}\ell - \int_{\gamma_g} (0,Q)\,\mathrm{d}\ell - \left(\int_{\gamma_g} (P,0)\,\mathrm{d}\ell - \int_{\gamma_f} (P,0)\,\mathrm{d}\ell\right) \\
&= \int_{\delta_b} F\,\mathrm{d}\ell - \int_{\delta_a} F\,\mathrm{d}\ell + \int_{\gamma_f} (0,Q)\,\mathrm{d}\ell - \int_{\gamma_g} (0,Q)\,\mathrm{d}\ell - \int_{\gamma_g} (P,0)\,\mathrm{d}\ell + \int_{\gamma_f} (P,0)\,\mathrm{d}\ell \\
&= \int_{\delta_b} F\,\mathrm{d}\ell - \int_{\delta_a} F\,\mathrm{d}\ell + \int_{\gamma_f} ((0,Q)+(P,0))\,\mathrm{d}\ell - \int_{\gamma_g} ((0,Q)+(P,0)\,\mathrm{d}\ell) \\
&= \int_{\delta_b} F\,\mathrm{d}\ell - \int_{\delta_a} F\,\mathrm{d}\ell + \int_{\gamma_f} F\,\mathrm{d}\ell - \int_{\gamma_g} F\,\mathrm{d}\ell
\end{aligned}$$

Where the last equal is since $F = (P,Q)$. □

Question 5

Let $\Omega = (a,b) \times (c,d) \subseteq \mathbb{R}^2$ be an open rectangle.

Subquestion 1

We will prove that Ω is not a smooth domain.

Proof: We need to show that $\partial\Omega$ is not a smooth curve.
The boundary $\partial\Omega$ contains the four line paths

$$\begin{cases} x = a & y \in [c,d] \\ x = b & y \in [c,d] \\ y = c & x \in [a,b] \\ y = d & x \in [a,b] \end{cases}$$

These line paths meet at the corner points $(a,c), (a,d), (b,c), (b,d)$.
Along each side the boundary is smooth but at the coreners the direction changes: if we take the point (a,c) the boundary is the union of two curves

$$\begin{cases} \gamma_1(t) = (t,c) & t > a \\ \gamma_2(t) = (a,s) & s > c \end{cases}$$

Obviously, $\gamma_1'(t) = (1,0) \neq (0,1) = \gamma_2'(s)$ hence there is no unique tangent vector the boundary is not smooth. $\square$

Subquestion 2

We will formulate and prove Green's theorem for rectangles using question 4.

Proof: Let $R = [a,b] \times [c,d]$ a rectangle and $F = (P,Q)$ a smooth vector field, then

$$\int_R \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \,\mathrm{d}x\,\mathrm{d}y = \oint_{\partial R} \vec{F}\,\mathrm{d}\vec{\ell}$$

We denote $f(x) = c, g(x) = d, A = R$ and ∂R is a closed curve consisting of four straight line segments described as before:

$$\begin{cases} \gamma_c(t) = (t,c) & t \in [a,b] \\ \gamma_d(t) = (t,d) & t \in [a,b] \\ \delta_b(t) = (b,t) & t \in [c,d] \\ \delta_a(t) = (a,t) & t \in [c,d] \end{cases}$$

Where $\gamma_c(t), \delta_b$ are oriented counterclockwise path, $\gamma_d(t), \delta_a(t)$ are opposite to the counterclockwise path which means that this path is $-\gamma_d, -\delta_a$ hence the total line integral over the positively oriented boundary ∂R is the sum of these four segments:

$$\oint_{\partial R} \vec{F}\,\mathrm{d}\vec{\ell} = \int_{\gamma_c} F\,\mathrm{d}\ell + \int_{\delta_b} F\,\mathrm{d}\ell - \int_{\gamma_d} F\,\mathrm{d}\ell - \int_{\delta_a} F\,\mathrm{d}\ell$$

From the conclusion in question 4 (with A as R and f, g as c, d)

$$\int_R \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \,\mathrm{d}x\,\mathrm{d}y = \int_{\delta_b} F\,\mathrm{d}\ell - \int_{\gamma_d} F\,\mathrm{d}\ell - \int_{\delta_a} F\,\mathrm{d}\ell + \int_{\gamma_c} F\,\mathrm{d}\ell$$

$$\underset{\text{Rearrange terms}}{\Longrightarrow} \int_R \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \,\mathrm{d}x\,\mathrm{d}y = \oint_{\partial R} \vec{F}\,\mathrm{d}\vec{\ell}$$

$\square$